

Literature Status: The Extreme-Point Problem In arXiv:2302.13951 Is Answered By arXiv:2412.04312

Automatic math-discovery smoke test

June 29, 2026

Status

This is a scoped **literature_already_answered** packet. It records that one open-problem signal in Aliaga–Pernecka–Smith, *Convex integrals of molecules in Lipschitz-free spaces*, arXiv:2302.13951 [1], is answered by the later paper Aliaga–Pernecka–Smith, *A solution to the extreme point problem and other applications of Choquet theory to Lipschitz-free spaces*, arXiv:2412.04312 [2].

Source Open Problem Signal

In Section 6 of arXiv:2302.13951, titled “Applications to extremal structure”, the source paper says that the main open problem in this direction is the characterization of the extreme points of the unit ball of $\mathcal{F}(M)$. More specifically, it identifies the unproved conjecture that all extreme points must be elementary molecules. The same paragraph notes that the conjecture was known for proper metric spaces and subsets of $\mathbb{R}$ -trees, and then proves the claim for extreme points representable as convex integrals of molecules.

Supporting Answer

The later paper arXiv:2412.04312 explicitly frames the same problem as the long-standing extreme point problem for Lipschitz-free spaces. Its Section 4, “The extreme point problem”, states that the problem is to determine whether all extreme points must be molecules, and then says that Theorem 4.1 provides a definitive answer.

Theorem 4.1 in arXiv:2412.04312. Let M be a complete metric space. Then $\text{ext}B_{\mathcal{F}(M)} \subset \text{Mol}(M)$. Specifically, the extreme points of $B_{\mathcal{F}(M)}$ are exactly the molecules m_{xy} , $x \neq y$, such that

$$d(x, p) + d(p, y) > d(x, y) \quad (p \in M \setminus \{x, y\}).$$

Conclusion

This answers the extreme-point conjecture/open problem mentioned in Section 6 of arXiv:2302.13951, for the same complete-metric-space setting used in the source paper.

Scope Limitations

This packet is deliberately narrow. It does not claim that arXiv:2412.04312 answers the source paper’s explicit Question in Section 3 about rewriting a convex series of molecules so that no

first coordinate is a second coordinate. It also does not claim to answer the explicit Question in Section 4 asking whether the isometric-copy hypothesis in Theorem 4.1 of arXiv:2302.13951 can be replaced by a bi-Lipschitz-copy hypothesis. In fact, arXiv:2412.04312 recasts a closely related curve-fragment question as an open question.

Evidence Checked

The local source TeX for arXiv:2302.13951 was inspected around the source open-problem signal and the two explicit questions. The local source TeX for arXiv:2412.04312 was inspected around its abstract, introduction, and Theorem `th:extreme`. A web check on June 29, 2026 found the arXiv pages for both the source paper and the supporting paper. The run indexes had an adjacent literature-status packet for another source paper and arXiv:2412.04312, but no exact packet for arXiv:2302.13951.

References

- [1] Ramón J. Aliaga, Eva Pernecká, and Richard J. Smith, *Convex integrals of molecules in Lipschitz-free spaces*, arXiv:2302.13951, 2023.
- [2] Ramón J. Aliaga, Eva Pernecká, and Richard J. Smith, *A solution to the extreme point problem and other applications of Choquet theory to Lipschitz-free spaces*, arXiv:2412.04312, 2024.